

EXERCISE 4

Intelligent data analysis DV1597

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The exercise should be performed **individually**, i.e. no group cooperation. Please hand in your written as a **PDF** file via Canvas. The report should be in **English** and include your name and the answers to the task below. The exercise is grade with G/Ux/U.

The Task

You have a data set of different faces (5 persons) in the folder **People**. Your task is to write a code that automatically recognizes the faces in the folder **Test** (you have 5 faces) using the Euclidean distance metric. The final result should assign the 5 faces to their correct names based on the smallest Euclidean distance as shown in Table 1.

Table 1: Calculated distances of images in “Test” to the means of people. Prediction is the one with the least distance (diagonal direction).

Test Image	Andreas	Marie	Mikael	Stefan	Ulf	True Label
1.png	6.4041	77.653	9.9427	77.204	67.354	Andreas
2.png	58.531	16.855	54.847	31.45	18.162	Marie
3.png	2.5797	70.149	2.5407	70.516	60.173	Mikael
4.png	66.199	21.268	63.298	4.9464	12.029	Stefan
5.png	63.061	13.526	59.847	11.263	2.7807	Ulf

How it Works

Step 1

First, calculate the statistical mean and standard deviation of all images in each of the 5 folders (**Andreas**, **Marie**, **Mikael**, **Stefan**, **Ulf**). Let p denote a person folder (i.e., one of the five folders), and j be the index representing an image in the folder. Then calculate the average of the means and the average of the standard deviations for each person p_j : $\bar{\mu}_{pj}$, $\bar{\sigma}_{pj}$

$\bar{\mu}_{pj}$	$\bar{\sigma}_{pj}$	Person (p_j)
103.9146	43.4088	Andreas
170.1453	69.7032	Marie
106.6726	45.9677	Mikael
174.5176	50.2704	Stefan
163.0294	58.2969	Ulf

Step 2

Calculate the mean and standard deviation of each of the images in the folder **Test**.

Step 3

Calculate the Euclidean distances of each image in the folder **Test** to the mean of the individuals using:

$$d_{tkpj} = \sqrt{(\mu_{tk} - \bar{\mu}_{pj})^2 + (\sigma_{tk} - \bar{\sigma}_{pj})^2} \quad (1)$$

Where:

- t denotes test folder
- p denotes person folder
- k, j are the image indices

For example, when $k = 1$ and $j = 1$, we see that: $d_{t1p1} = 6.4041$, and when $k = 2$ and $j = 4$, we see that: $d_{t2p4} = 31.4452$; as shown in Table 1.

Deliverables

The report should include the following items generated by your code:

1. A scatter plot showing the mean and standard deviation of all 9 images per person. For reference, an example of the expected output is shown in Figure 1 (please note that this is a sample illustration and not the actual result).
2. The distance table as shown in Table 1.
3. A discussion of patterns observed in the scatter plot. Was it easier to spot patterns in the plot than in the table?

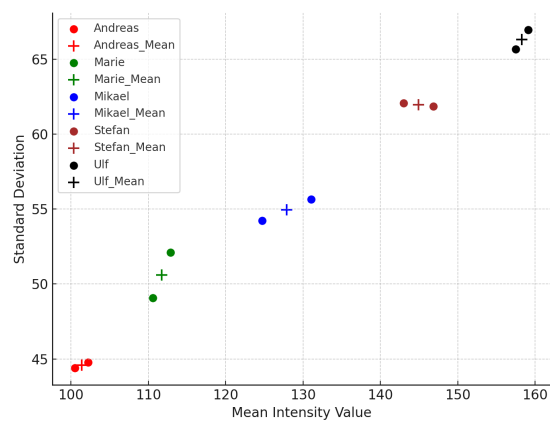


Figure 1: Example of a scatter plot showing clustering of images by person.